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handling_missind_data.py

```
1 import pandas as pd
2 import numpy as np
3 data = {
4     'Name': ['Alice', 'Bob', 'Charlie', 'David'],
5     'Age': [25, np.nan, 30, 22],
6     'Salary': [50000, 60000, np.nan, 45000]
7 }
8 df = pd.DataFrame(data)
9 # 1. Check for missing values
10 print("Missing count:\n", df.isnull().sum())
11 # 2. Fill missing Age with the column median
12 df['Age'] = df['Age'].fillna(df['Age'].median())
13 # 3. Drop rows where Salary is missing
14 df = df.dropna(subset=['Salary'])
15 print("\nCleaned DataFrame:\n", df)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\FDS> cd 'd:\FDS'; & 'C:\Users\VACER\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\VACER\.vscode\extensions\ms-python.debugpy-2026.6.0-wi
n32-x64\bundled\libs\debugpy\launcher' '54466' '-' 'D:\FDS\handling_missind_data.py'

Missing count:
Name 0
Age 1
Salary 1
dtype: int64

Cleaned DataFrame:
Name Age Salary
0 Alice 25.0 50000.0
1 Bob 25.0 60000.0
3 David 22.0 45000.0
PS D:\FDS>]

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filtering_rows.py

```
1 import pandas as pd
2
3 data = {
4     'Employee': ['Amy', 'John', 'Raj', 'Kevin', 'Sora'],
5     'Department': ['HR', 'IT', 'IT', 'Sales', 'IT'],
6     'Experience': [3, 7, 1, 5, 6]
7 }
8
9 df = pd.DataFrame(data)
10
11 # filter for employees in IT with more than 5 years of experience
12 filtered_df = df[(df['Department'] == 'IT') & (df['Experience'] > 5)]
13
14 print(filtered_df)
```

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n32-x64\bundled\libs\debugpy\launcher' '50913' '-' 'D:\FDS\filtering_rows.py'

```
1 John IT 7
4 Sora IT 6
PS D:\FDS> ]
```

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aggregating_data.py

```
1 import pandas as pd
2
3 data = {
4     'Store': ['North', 'South', 'North', 'West', 'South', 'West'],
5     'Sales': [1200, 1500, 900, 1100, 1700, 1300],
6     'Employees': [5, 7, 5, 4, 8, 4]
7 }
8
9 df = pd.DataFrame(data)
10
11 # Group by Store and calculate total sales and average employees
12 summary = df.groupby('Store').agg(
13     'Sales': 'sum',
14     'Employees': 'mean'
15 ).reset_index()
16
17 print(summary)
```

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	Store	Sales	Employees
0	North	2100	5.0
1	South	3200	7.5
2	West	2400	4.0

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describe_function.py

```
1 import pandas as pd
2
3 data = {
4     'Age': [25, 30, 35, 40, 45],
5     'Salary': [50000, 60000, 75000, 90000, 110000],
6     'Department': ['HR', 'IT', 'IT', 'Marketing', 'HR']
7 }
8
9 df = pd.DataFrame(data)
10
11 print("--- Numeric Summary ---")
12 print(df.describe())
13
14 print("\n--- Categorical Summary ---")
15 print(df.describe(include='object'))
```

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--- Numeric Summary ---

	Age	Salary
count	5.000000	5.000000
mean	35.000000	77000.000000
std	7.905694	23874.672773
min	25.000000	50000.000000
25%	30.000000	60000.000000
50%	35.000000	75000.000000
75%	40.000000	90000.000000
max	45.000000	110000.000000

--- Categorical Summary ---

D:\FDS\describe_function.py:19: PandasWarning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to 'include' to select them, or to 'exclude' to remove them and silence this warning. See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.

	Department
count	5
unique	3
top	HR
freq	2

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